

SURVEY PAPER

**THE M-WRIGHT FUNCTION
AS A GENERALIZATION OF THE GAUSSIAN DENSITY
FOR FRACTIONAL DIFFUSION PROCESSES**

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*Dedicated to Professor Francesco Mainardi
on the occasion of his 70th anniversary*

Abstract

The leading role of a special function of the Wright-type, referred to as M-Wright or Mainardi function, within a parametric class of self-similar stochastic processes with stationary increments, is surveyed. This class of processes, known as generalized grey Brownian motion, provides models for both fast and slow anomalous diffusion. In view of a subordination-type formula involving M-Wright functions, these processes emerge to have all finite moments and be uniquely defined by their mean and auto-covariance structure like Gaussian processes. The corresponding master equation is shown to be a fractional differential equation in the Erdélyi–Kober sense and the diffusive process is named Erdélyi–Kober fractional diffusion. In Appendix, an historical overview on the M-Wright function is reported.

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1. Introduction

The observation in nature of the *anomalous diffusion* has been definitively established. Some noteworthy examples are shown in References [8, 10, 25, 43, 55].

The label *anomalous diffusion* is termed in opposition to the *normal diffusion*, where the adjective *normal* has the double meaning to highlight a Gaussian based diffusion (because of the correspondence between the Normal and the Gaussian density) and of a typical and usual diffusion process conformed to the standard. Actually, the label *anomalous diffusion* is assigned to processes whose particle variance does not grow linearly in time, which indeed is the main characteristic of the classic, and then normal, Gaussian based Brownian motion. A number of approaches for explaining the anomalous diffusion have been introduced in literature. A successful one is that based on evolution equations of fractional order and the name *fractional diffusion* follows. This approach has been maintained by analytical methods, e.g., [6, 7, 37], by random walks, e.g., [15, 20, 21, 22], by the continuous time random walk (CTRW) through a well-scaled passage to the diffusion limit, e.g., [16, 17, 18, 23], by the parametric subordination, e.g., [19], and by the inverse stable subordination, e.g., [1, 42].

The M-Wright function, which is a transcendental function of the Wright-type, was originally introduced by by Professor Francesco Mainardi at the beginning of the 1990's as the fundamental solution (Green function) of the initial value problem for the time-fractional diffusion-wave equation. When the order of the time derivative is less or equal to 1, this Green function can be interpreted as the probability density function (*pdf*) evolving in time of the particle position for a relevant class of self-similar stochastic processes, which are generally referred to as time-fractional diffusion processes. Furthermore, when these (in general non-Markovian) diffusion stochastic processes are properly characterized with stationary increments, the M-Wright function is shown to play the same key role as the Gaussian density for the standard and the fractional Brownian motions. These stochastic processes, whose class is named *generalized grey Brownian motion* (ggBm), are more general than the classical standard and fractional Brownian motions and are suitable to model anomalous diffusion [38, 46, 47, 48]. The properties of the corresponding Master Equation (ME) lead to name such diffusion as Erdélyi-Kober fractional diffusion [50, 51, 52]. Actually, the M-Wright function was also used as a non-Markovian model to generalize the evolution in time of the radius of a premixed flame ball in fractional diffusive media [49, 50] and it has emerged to be related to the quadratic variation for compound renewal processes [56] whose convergence is important for stochastic integrals driven by a CTRW [11].

Due to this distinguished role, the M-Wright function deserved its own proper analysis [3, 4, 5, 38]. Mainardi, in his first studies on the time-fractional diffusion equation in the Caputo sense, introduced the two Wright-type functions $F_\nu(z)$ and $M_\nu(z)$, $0 < \nu < 1$, for two different boundary value problems, where ν is the half of the fractional order of the time derivative [31, 34].

As a matter of fact these functions $F_\nu(z)$ and $M_\nu(z)$ are particular cases of the Wright function of the second kind $W_{\lambda,\mu}(z)$, i.e.,

$$W_{\lambda,\mu}(z) = \sum_{n=0}^{\infty} \frac{z^n}{n! \Gamma(\lambda n + \mu)}, \quad \lambda > -1, \quad \mu \in C, \quad z \in C, \quad (1.1)$$

by setting $\lambda = -\nu$ and $\mu = 0$ or $\mu = 1 - \nu$, respectively. Namely,

$$F_\nu(z) = W_{-\nu,0}(-z), \quad M_\nu(z) = W_{-\nu,1-\nu}(-z), \quad 0 < \nu < 1, \quad (1.2)$$

and they are inter-related through $F_\nu(z) = \nu z M_\nu(z)$. Their series representation are:

$$F_\nu(z) = \sum_{n=1}^{\infty} \frac{(-z)^n}{n! \Gamma(-\nu n)} = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{(-z)^{n-1}}{n!} \Gamma(\nu n + 1) \sin(\pi \nu n), \quad (1.3)$$

$$M_\nu(z) = \sum_{n=0}^{\infty} \frac{(-z)^n}{n! \Gamma[-\nu n + (1 - \nu)]} = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{(-z)^{n-1}}{(n-1)!} \Gamma(\nu n) \sin(\pi \nu n). \quad (1.4)$$

The noteworthy particular case is

$$M_{1/2}(z) = \frac{1}{\sqrt{\pi}} \exp(-z^2/4), \quad (1.5)$$

and, in spite of the fact that for some particular properties $M_\nu(z)$ can be considered a *generalized hyper-Airy function* [41], in view of its leading role in diffusive stochastic processes, it is now considered as a natural (fractional) generalization of the Gaussian density.

Further properties of the M-Wright function can be found in the book of Mainardi [35], especially in Chapter 6 and Appendix F.

In the literature, the M-Wright function $M_\nu(z)$ is also referred to as the Mainardi function. This name was originally introduced in the community of fractional analysts by Podlubny [54] and now adopted after Baleanu [2] also in the physics community, see, e.g. [44].

The rest of the paper is organized as follows. In Section 2 a composition formula of the subordination-type for the M-Wright function is re-derived and its relationship with the Gaussian density is highlighted. In Section 3 the generalized ME for self-similar modelling of fractional diffusion is considered and its Green function based on the M-Wright function shown. Moreover, the underlying ggBm and the Erdélyi-Kober fractional diffusion

are discussed. In Section 4 conclusions are given and in Appendix the story is told of the hard path accomplished by the M-Wright function to appear in scientific peer-reviewed literature.

2. Subordination-type formula for the M-Wright function

The subordination-type formulae are special integral representations of positive functions that can be interpreted as *pdf* associated to stochastic processes. For self-similar processes, two different stochastic interpretations of subordination formulae are given in literature and these interpretations have been recently unified [53].

Before to show the subordination-type formula for the M-Wright function, originally derived in Reference [40], the definition of the Mellin transform of a sufficiently well-behaved function $f(r)$, $r \in R^+$, is here reminded to be

$$\mathcal{M}\{f(r); s\} = f^*(s) = \int_0^{+\infty} f(r) x^{s-1} dr, \quad s \in C, \quad (2.1)$$

when the integral converges. According to the most usual approach suitable for applied scientists, it is assumed $f(r) \in L_{loc}(R^+)$. Integral (2.1) defines the Mellin transform in a vertical strip in the s -plane whose boundaries are determined by the analytic structure of $f(r)$ as $r \rightarrow 0^+$ and $r \rightarrow +\infty$. When

$$f(r) = \begin{cases} O(r^{-\gamma_1-\epsilon}); & r \rightarrow 0^+ \\ O(r^{-\gamma_2-\epsilon}); & r \rightarrow +\infty, \end{cases} \quad (2.2)$$

then for every (small) $\epsilon > 0$ and $\gamma_1 < \gamma_2$, the integral (2.1) converges absolutely and defines an analytic function in the strip $\gamma_1 < \text{Re}(s) < \gamma_2$.

The inversion formula for (2.1) follows directly from the corresponding inversion formula for the bilateral Laplace transform and it results to be

$$\mathcal{M}^{-1}\{f^*(s); r\} = f(r) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} f^*(s) r^{-s} ds, \quad \gamma_1 < \gamma < \gamma_2, \quad (2.3)$$

at all points $r \geq 0$ where $f(r)$ is continuous.

Here it is reminded the *Mellin convolution* formula

$$h(r) = \int_0^\infty f\left(\frac{r}{\xi}\right) g(\xi) \frac{d\xi}{\xi} \xleftrightarrow{\mathcal{M}} f^*(s) g^*(s) = h^*(s), \quad (2.4)$$

which is fundamental for the rest of the section.

Combining series representation (1.4) and the inversion formula for the Mellin transform (2.3), by reminding the Residue Theorem and that the residue of the Gamma function at $-n$ is $(-1)^n/n!$ for $n = 0, 1, 2, \dots$, it

follows that

$$M_\nu(r) \xleftrightarrow{\mathcal{M}} \frac{\Gamma[1+(s-1)]}{\Gamma[1+\nu(s-1)]}, \quad 0 < \nu < 1. \quad (2.5)$$

When the evolution in time is considered according to a law of self-similarity, then it results

$$M_\nu(r; t) = t^{-\nu} M_\nu\left(\frac{r}{t^\nu}\right), \quad (2.6)$$

hence the following theorem holds [40].

THEOREM. *Let $M_\nu(r; t)$, $M_\eta(r; t)$ and $M_\beta(r; t)$ be M -Wright functions of orders $\nu, \eta, \beta \in (0, 1)$ respectively, then the following subordination-type formula holds true for $0 < r < \infty$,*

$$M_\nu(r; t) = \int_0^\infty M_\eta(r; \tau) M_\beta(\tau; t) d\tau, \quad \text{with } \nu = \eta\beta. \quad (2.7)$$

Because of scaling property (2.6), formula (2.7) can be recasted as

$$t^{-\nu} M_\nu\left(\frac{r}{t^\nu}\right) = \int_0^\infty \tau^{-\eta} M_\eta\left(\frac{r}{\tau^\eta}\right) t^{-\beta} M_\beta\left(\frac{\tau}{t^\beta}\right) d\tau. \quad (2.8)$$

Formula (2.8) can be proven by using (2.5) that, after some algebra, gives

$$\mathcal{M}\{t^{-\nu} M_\nu\left(\frac{r}{t^\nu}\right); s\} = \mathcal{M}\{b c r^a M_\beta(c r^b); s\} \mathcal{M}\{M_\eta(r); s\}, \quad (2.9)$$

provided that

$$a = \frac{1}{\eta} - 1, \quad b = \frac{1}{\eta}, \quad c = t^{-\nu/\eta}, \quad (2.10)$$

and

$$\nu = \beta\eta. \quad (2.11)$$

Recalling Mellin convolution formula (2.4), from (2.9)-(2.11) it follows that

$$t^{-\nu} M_\nu\left(\frac{r}{t^\nu}\right) = \int_0^\infty \frac{1}{\eta} \xi^{1/\eta-1} M_\eta\left(\frac{r}{\xi}\right) t^{-\beta} M_\beta\left(\frac{\xi^{1/\eta}}{t^\beta}\right) \frac{d\xi}{\xi}, \quad (2.12)$$

and, by replacing $\xi \rightarrow \tau^\eta$, formula (2.8) is obtained.

When $\nu = \beta/2$ and $\eta = 1/2$, by using noteworthy property (1.5), the subordination-type formula (2.7) gives the following valuable relation

$$M_{\beta/2}(r; t) = 2 \int_0^\infty \frac{1}{\sqrt{4\pi\tau}} e^{-r^2/(4\tau)} M_\beta(\tau; t) d\tau. \quad (2.13)$$

3. Self-similar non-Markovian time-fractional diffusion

3.1. Generalized master equation

A general formulation for the temporal evolution of the spatial probability density $P(x; t)$ of a non-Markovian diffusive process is

$$\frac{\partial P}{\partial t} = \int_0^t K(x, t - \tau) \frac{\partial^2 P(x; \tau)}{\partial x^2} d\tau, \quad (3.1)$$

where $K(x, t)$ denotes a memory kernel. It is worth to remark that until the memory kernel is not trivial (i.e., $K(x, t) = \delta(t)$) the *pdf* $P(x; t)$ is not in general a Gaussian *pdf*. Equation (3.1) can be further generalized assuming a “time-stretching” described by a smooth and increasing function $g(t)$, with $g(0) = 0$, so that $P(x; g(t)) = P_*(x; t)$ and then

$$\frac{\partial P_*}{\partial t} = \frac{dg}{dt} \int_0^t K[x, g(t) - g(\tau)] \frac{\partial^2 P_*}{\partial x^2} \frac{dg}{d\tau} d\tau. \quad (3.2)$$

Choosing a power memory kernel of the form

$$K(t) = \frac{t_+^{-\mu-1}}{\Gamma(-\mu)}, \quad 0 < \mu < 1, \quad (3.3)$$

where the suffix $_+$ is just denoting that the function is vanishing for $t < 0$, guarantees that P_* be a *pdf* and that the process be self-similar [48]. Moreover, by assuming the following “time-stretching” function

$$g(t) = t^{\alpha/\beta}, \quad 0 < \alpha < 2, \quad 0 < \beta \leq 1, \quad (3.4)$$

equation (3.2) becomes

$$\begin{aligned} \frac{\partial P_*}{\partial t} &= \frac{\alpha}{\beta} t^{\alpha/\beta-1} \int_{0-}^{t+} \frac{(t^{\alpha/\beta} - \tau^{\alpha/\beta})^{-\mu-1}}{\Gamma(-\mu)} \frac{\partial^2 P_*}{\partial x^2} \frac{\alpha}{\beta} \tau^{\alpha/\beta-1} d\tau \\ &= \frac{\alpha}{\beta} t^{\alpha/\beta-1} D_{t^{\alpha/\beta}}^{\mu} \frac{\partial^2 P_*}{\partial x^2}, \end{aligned} \quad (3.5)$$

where D_t^{μ} is the Riemann–Liouville fractional differential operator of order μ in its formal definition according to Reference [14, Eq. (1.34)]. The operator D_t^{μ} is obtained by using the representation of the generalized derivative of order n of the Dirac delta distribution: $\delta^{(n)}(t) = \frac{t_+^{-n-1}}{\Gamma(-n)}$, $n=0, 1, 2, \dots$, with proper interpretation of the quotient as a limit if $t=0$. For more details the reader is referred to Gorenflo and Mainardi [14]. By setting $\mu = 1 - \beta$, equation (3.5) corresponds to the stretched time-fractional diffusion equation [38, Eq. (5.19)] and it is the ME corresponding

to the following integral evolution equation

$$P_*(x; t) = P_{*0}(x) + \frac{1}{\Gamma(\beta)} \frac{\alpha}{\beta} \int_0^t \tau^{\alpha/\beta-1} (t^{\alpha/\beta} - \tau^{\alpha/\beta})^{\beta-1} \frac{\partial^2 P_*(x; \tau)}{\partial x^2} d\tau, \quad (3.6)$$

that is the integral evolution equation of the ggBm originally introduced by Mura [45] and later discussed in a number of papers [38, 46, 47, 48]. Actually, it was also recognized that the use of the Riemann–Liouville fractional operator with a stretched time variable is an abuse of notation, it being an Erdélyi–Kober type operator [50, 51, 52].

Special cases of ME (3.18) are straightforwardly obtained [51]. In particular, it reduces to the time-fractional diffusion if $\alpha = \beta < 1$, to the time-stretched Gaussian diffusion if $\alpha \neq 1$ and $\beta = 1$, and finally to the standard Gaussian diffusion if $\alpha = \beta = 1$.

The reduction to the time-fractional diffusion when $\alpha = \beta = 1$ can be used to compute the Green function. In fact, in this case the “time-stretching” disappears and it is well known from pioneering papers by Mainardi [33, 34] (see Appendix) that the Green function is

$$P(x; t) = \frac{1}{2} \frac{1}{t^{\beta/2}} M_{\beta/2} \left(\frac{|x|}{t^{\beta/2}} \right), \quad 0 < \beta < 1. \quad (3.7)$$

Hence, applying the time stretching $t \rightarrow t^{\alpha/\beta}$ to (3.7), the Green function corresponding to (3.16, 3.18) is [45, 46, 47, 48]

$$P_*(x; t) = \frac{1}{2} \frac{1}{t^{\alpha/2}} M_{\beta/2} \left(\frac{|x|}{t^{\alpha/2}} \right). \quad (3.8)$$

This *pdf* has all finite moments because of its exponential asymptotic decay [35, 37] and it can describe both slow and fast anomalous diffusion. In fact, its variance turns out to be

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} x^2 P_*(x; t) dx = \frac{2}{\Gamma(\beta + 1)} t^\alpha, \quad (3.9)$$

and the resulting process is self-similar with Hurst exponent $H = \alpha/2$. The variance law is thus consistent with slow diffusion when $0 < \alpha < 1$ and fast diffusion when $1 < \alpha \leq 2$. However, it is worth to be remarked also that a linear variance growing is possible, even with non-Gaussian *pdf*, when $\beta \neq \alpha = 1$.

3.2. The generalized grey Brownian motion

The Green function (3.8) of the generalized integral ME (3.6) can be interpreted as the *pdf* of a diffusing particle which evolves in time driven by a self-similar stochastic process. However, it is not possible to generally define a *unique* self-similar stochastic process by knowing solely the one-point *pdf*, the multi-point *pdf* is required [48].

In fact, it is always possible to define an equivalence class of stochastic processes with the same marginal *pdf* and all these processes provide suitable stochastic representations for the one-point ME. Hence, additional requirements are needed.

Following Mura and co-authors [45, 46, 47, 48], when the stationarity of increments is taken into account as additional requirement, this leads to a class of stochastic processes $\{B_{\alpha,\beta}(t), t \geq 0\}$ called “generalized” grey Brownian motion, which, *by construction*, is made up of self-similar processes with stationary increments and Hurst exponent $H = \alpha/2$. Thus $\{B_{\alpha,\beta}(t), t \geq 0\}$ is a special class of H -self-similar-stationary-increments (H-sssi) processes which provide non-Markovian stochastic models for anomalous diffusion, both of slow type ($0 < \alpha < 1$) and fast type ($1 < \alpha < 2$). However, it is worth noting to be remarked also that a linear variance growing is possible when $\beta \neq \alpha = 1$, in this case the increments are uncorrelated, but in general not independent, and the *pdf* is not Gaussian.

The ggBm generalizes some well known processes, so that it defines an interesting general theoretical framework. The fractional Brownian motion (fBm) appears for $\beta = 1$; the grey Brownian motion (gBm), defined by Schneider [58, 59], when $0 < \alpha = \beta < 1$, and the standard Brownian motion (Bm) is recovered by setting $\alpha = \beta = 1$. It is worth to remark that only in the particular case of the Bm the corresponding process turns out to be Markovian.

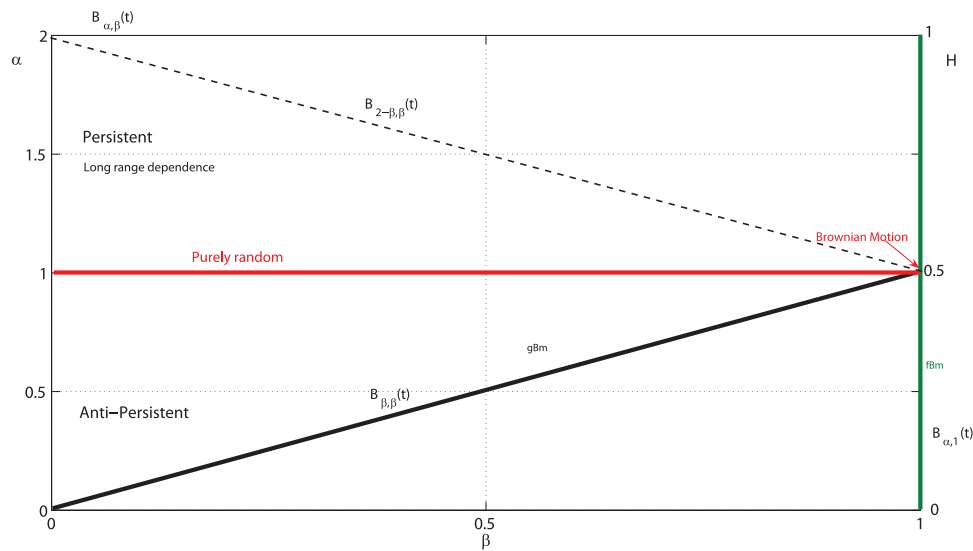


Figure 1: Element diagram of the parametric class of the generalized grey Brownian motion.

In Figure 1, a diagram is shown which allows to identify the elements of the ggBm class. The top region $1 < \alpha < 2$ corresponds to the domain of fast diffusion with *long-range dependence*. In this domain the increments of the process $B_{\alpha,\beta}(t)$ are positively correlated, so that the trajectories tend to be more regular (*persistent*). It should be noted that long-range dependence is associated to a non-Markovian process which exhibits long-memory properties. The horizontal line $\alpha = 1$ represents processes with purely random, i.e., uncorrelated, increments, which model various phenomena of normal diffusion. For $\alpha = \beta = 1$ the standard Bm follows. The fBm is identified by the vertical line $\beta = 1$. The bottom region $0 < \alpha < 1$ corresponds to the domain of slow diffusion. The increments of the corresponding process $B_{\alpha,\beta}(t)$ turn out to be negatively correlated and this implies that the trajectories are very “zigzagging” (*anti-persistent*) and the increments generate a stationary process which does not exhibit long-range dependence. Finally, the lower diagonal line ($\alpha = \beta$) represents the Schneider gBm whereas the upper diagonal line indicates its “conjugated” process.

The ggBm can be defined considering the Kolmogorov extension theorem and the properties of the M-Wright function. In fact, the ggBm $B_{\alpha,\beta}(t)$ is a stochastic process defined in a certain probability space such that its finite-dimensional distributions are given by [47]

$$f_{\alpha,\beta}(x_1, x_2, \dots, x_n; \gamma_{\alpha,\beta}) = \frac{(2\pi)^{-\frac{n-1}{2}}}{\sqrt{2\Gamma(1+\beta)^n \det \gamma_{\alpha,\beta}}} \times \int_0^\infty \frac{1}{\tau^{n/2}} M_{1/2}\left(\frac{\xi}{\tau^{1/2}}\right) M_\beta(\tau) d\tau, \quad (3.10)$$

where

$$\xi = \left(2\Gamma(1+\beta)^{-1} \sum_{i,j=1}^n x_i \gamma_{\alpha,\beta}^{-1}(t_i, t_j) x_j \right)^{1/2}, \quad (3.11)$$

and the covariance matrix is

$$\gamma_{\alpha,\beta}(t_i, t_j) = \frac{1}{2\Gamma(1+\beta)} (t_i^\alpha + t_j^\alpha - |t_i - t_j|^\alpha), \quad i, j = 1, \dots, n. \quad (3.12)$$

This covariance matrix characterizes the typical dependence structure of a self-similar process with stationary increments and Hurst exponent $H = \alpha/2$, [61]. For the one-point case, i.e., $n = 1$, by using subordination-type formula (2.13), formula (3.10) reduces to

$$\begin{aligned} f_{\alpha,\beta}(x; t) &= \frac{1}{\sqrt{4t^\alpha}} \int_0^\infty M_{1/2}\left(|x|t^{-\alpha/2}; \tau\right) M_\beta(\tau; 1) d\tau \\ &= \frac{1}{2} t^{-\alpha/2} M_{\beta/2}(|x|t^{-\alpha/2}) = P_*(x; t). \end{aligned} \quad (3.13)$$

This means that the marginal *pdf* is indeed the solution of (3.6). Moreover, setting $\beta = 1$, Green function (3.10) reduces to the Gaussian distribution of the fBm

$$f_{\alpha,1}(x_1, x_2, \dots, x_n; \gamma_{\alpha,1}) = \frac{(2\pi)^{-\frac{n-1}{2}}}{\sqrt{2 \det \gamma_{\alpha}}} M_{1/2} \left(\left(2 \sum_{i,j=1}^n x_i \gamma_{\alpha}^{-1}(t_i, t_j) x_j \right)^{1/2} \right), \quad (3.14)$$

where $\gamma_{\alpha} \equiv \gamma_{\alpha,1}$, because it holds $M_1(\tau) = \delta(\tau - 1)$. Finally it reduces to the standard Gaussian *pdf* of Bm when $\alpha = 1$.

From definition used above, it clearly emerges that the ggBm is a natural generalization of the Gaussian process, which is recovered as special case. In fact, for fixed β , the ggBm is uniquely characterized through only its first and second moments [46, 47], which indeed is a remarkable property of the Gaussian process. Consequently, in the same sense, the M-Wright function is the natural generalization of the Gaussian density.

3.3. The Erdélyi–Kober fractional diffusion

It is well-known that there exists a relationship between the solutions of a certain class of integral equations that are used to model anomalous diffusion and stochastic processes. In this respect, the density function $P_*(x; t)$, which solves (3.6), can be understood as the marginal *pdf* of the ggBm [45, 46, 47, 51]. The ME corresponding to (3.6) is (3.5). But, an abuse of notation occurs in (3.5) because the Riemann–Liouville fractional differential operator is used with a stretched time variable.

Following [51], the integral in the non-Markovian kinetic equation (3.6) can be expressed in terms of an Erdélyi–Kober fractional integral operator $I_{\eta}^{\gamma, \mu}$ that, for a sufficiently well-behaved function $\varphi(t)$, is defined as (see for example, [24, Eq. (1.1.17)])

$$I_{\eta}^{\gamma, \mu} \varphi(t) = \frac{\eta}{\Gamma(\mu)} t^{-\eta(\mu+\gamma)} \int_0^t \tau^{\eta(\gamma+1)-1} (t^{\eta} - \tau^{\eta})^{\mu-1} \varphi(\tau) d\tau, \quad (3.15)$$

where $\mu > 0$, $\eta > 0$ and $\gamma \in \mathcal{R}$. Hence equation (3.6) can be re-written as [50, 51]

$$P_*(x; t) = P_{*0}(x) + t^{\alpha} \left[I_{\alpha/\beta}^{0, \beta} \frac{\partial^2 P_*}{\partial x^2} \right]. \quad (3.16)$$

Since the ggBm serves as a stochastic model for the anomalous diffusion, this leads to define the family of diffusive processes governed by the ggBm as *Erdélyi–Kober fractional diffusion* [51].

Due to the correspondence between (3.6) and (3.16), the correct expression for ME (3.5) is obtained by introducing the Erdélyi–Kober fractional differential operator $D_{\eta|t}^{\gamma, \mu}$ that is defined, for $n - 1 < \mu \leq n$, as [24, Eq. (1.5.19)]

$$D_{\eta|t}^{\gamma,\mu} \varphi(t) = \prod_{j=1}^n \left(\gamma + j + \frac{1}{\eta} t \frac{d}{dt} \right) (I_{\eta}^{\gamma+\mu, n-\mu} \varphi(t)). \quad (3.17)$$

Recently, the notions of Erdélyi–Kober fractional integrals and derivatives have been further extended [26, 27].

Finally, the ME of the ggBm, or Erdélyi–Kober fractional diffusion, is [51]

$$\frac{\partial P_*}{\partial t} = \frac{\alpha}{\beta} t^{\alpha-1} D_{\alpha/\beta|t}^{\beta-1, 1-\beta} \frac{\partial^2 P_*}{\partial x^2}. \quad (3.18)$$

4. Conclusions

In the present paper the leading role of the M-Wright function in fractional diffusion is considered. It is shown that this function is the natural generalization of the Gaussian density for non-Markovian fractional diffusion processes.

This important property mathematically follows from a strong relationship derived through an integral representation formula that can be read as a subordination-type formula. In particular, through the Kolmogorov extension theorem and this formula, the ggBm is defined and it provides an example of stochastic process characterized by only the mean and the autocovariance matrix, which is a valuable property of the Gaussian processes. Then the ggBm emerges to be a natural generalization of the Gaussian process, recovered as particular cases. The M-Wright function results to be the marginal *pdf* with all finite moments so that it can be interpreted as a natural generalization of the Gaussian density for non-Markovian fractional diffusion processes.

The ggBm is a parametric class of stochastic processes that provides models for both fast and slow anomalous diffusion as illustrated in the diagram of Figure 1. In this class are included the fractional Brownian motion, the Schneider grey Brownian motion and the standard Brownian motion. Since the ME of the ggBm is correctly understood in the sense of the Erdélyi–Kober fractional operators, the family of these diffusive processes can be named Erdélyi–Kober fractional diffusion.

Appendix: Historical notes on the M-Wright function

The first time that Professor Francesco Mainardi presented the two functions of the Wright-type $F_{\nu}(z)$ and $M_{\nu}(z)$ it was in the summer 1993, during the *IUTAM Symposium on Nonlinear Waves in Solids*, at the University of Victoria, Canada, 15–20 August 1993. Unfortunately, the symposium contribution appeared, as well as the others, solely as IUTAM Invited Symposium Paper Abstract in a special number of the *Applied Mechanics*

Reviews [28], apart from the four general lectures which were published as full papers. However, that first contribution was subsequently published as a proceeding paper [32] in 1995.

Actually, it was in his home-town Bologna, immediately after the summer, during the *7th Conference on Waves and Stability in Continuous Media* (WASCOM), Bologna, Italy, 4–7 October 1993, that Mainardi presented the M-Wright function with more details and plots. This communication [30] was published in the corresponding Proceedings book in 1994.

The subsequent year, Mainardi presented the M-Wright function in 4 international contexts.

At the *2nd Int. Sci. School-Seminar: Dynamics and Stochastic Wave Phenomena*, Nizhny Novgorod, Russia, 21–28 June 1994, but the contribution [31] was published in 1995. There the proceeding paper [41] and the classical paper in *Chaos, Solitons & Fractals* [33] were referred to as “*in press*”.

At the *Int. Summer School on Fractal and Hyperbolic Geometry*, Bordeaux, France, 3–8 July 1994, during which Mainardi met Professor Rudolf Gorenflo and they started their close and successful collaboration.

At the *12th IMACS World Congress*, Georgia Tech, Atlanta, USA, 11–15 July 1994, and the contribution [29] was published in the proceedings in the same year.

At the *1st Int. Workshop “Transform Methods and Special Functions”*, Sofia, 12–17 August 1994, and in the proceedings published the following year, [41]. During the presentation by Mainardi, Professor Bogoljub Stanković informed him that the proof that the radius of convergence of the power series (1.3) and (1.4) is infinite for $0 < \nu < 1$ [30] was already made just by Wright himself [62] in 1940, following his previous papers in 1930’s. Mainardi obtained such extension independently, being not aware of neither Wright 1940 paper [62] and Stanković 1970 paper [60], but solely of the Bateman Project [9, Vol. 3, Chapter 18], where, with reference to definition (1.1), the extension to $-1 < \lambda < 0$ was not considered and the condition $\lambda > 0$ was stated presumably for a misprint. Later, in the paper [36] devoted to the 80th birthday of Stanković, Mainardi took the occasion to renew his personal gratitude to him for this earlier information that has induced him to study the original papers by Wright and to work (also in collaboration) on the functions of the Wright-type, see, e.g., [12, 13, 39].

Moreover, by reading references in [41], the reader can learn the fact that a proceeding book was planned after the Bordeaux July 1994 conference in the journal *Chaos, Solitons & Fractals* and edited by A. Le Méhauté and A. Oustaloup. However, that proceedings never appeared

and Mainardi independently submitted his contribution [33] in December 1995 to the same journal, and it was published in 1996.

In the same 1996, Mainardi published in the journal *Applied Mathematics Letters* the other paper concerning the M-Wright function [34].

Up to now, beginning of February 2013 *, looking at the conservative citation database Web of Science, the papers [33] and [34] have 271 and 129 citations, respectively.

To conclude this historical note, it is reminded that the time-fractional diffusion equation was investigated via the Mellin transform and the Fox H -functions by Schneider & Wyss in 1989, in their pioneering paper [57]. However, Mainardi has simplified the approach by Schneider & Wyss by using the Laplace transform and noting in an explicit way that the fundamental solution can be expressed in terms of a special function of the Wright-type, a fact that was not known before. In addition, he has studied the analytical properties of this function and provided for the first time some significant plots. Unfortunately, due to some referees, who in those times were strongly opposed to fractional calculus, the final results have appeared in refereed international journals only some years later [33, 34].

Quoting from [35], as an instructive example of volatility of ideas in science as in fashion, the report of an anonymous referee of the twentieth century (June 1994) is here reproduced, clearly against the use of fractional calculus in mathematical physics:

This paper is of insufficient interest to publish as a Letter; I believe that it is sufficiently straightforward not to consider publishing it as a paper, either. The main drawback is that no application, either Physical or Mathematical, is really identified; what are these equations for? Anyone can write down some linear equations and then solve them, but that is not the point of doing Mathematical Physics. The paper should be rejected.

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